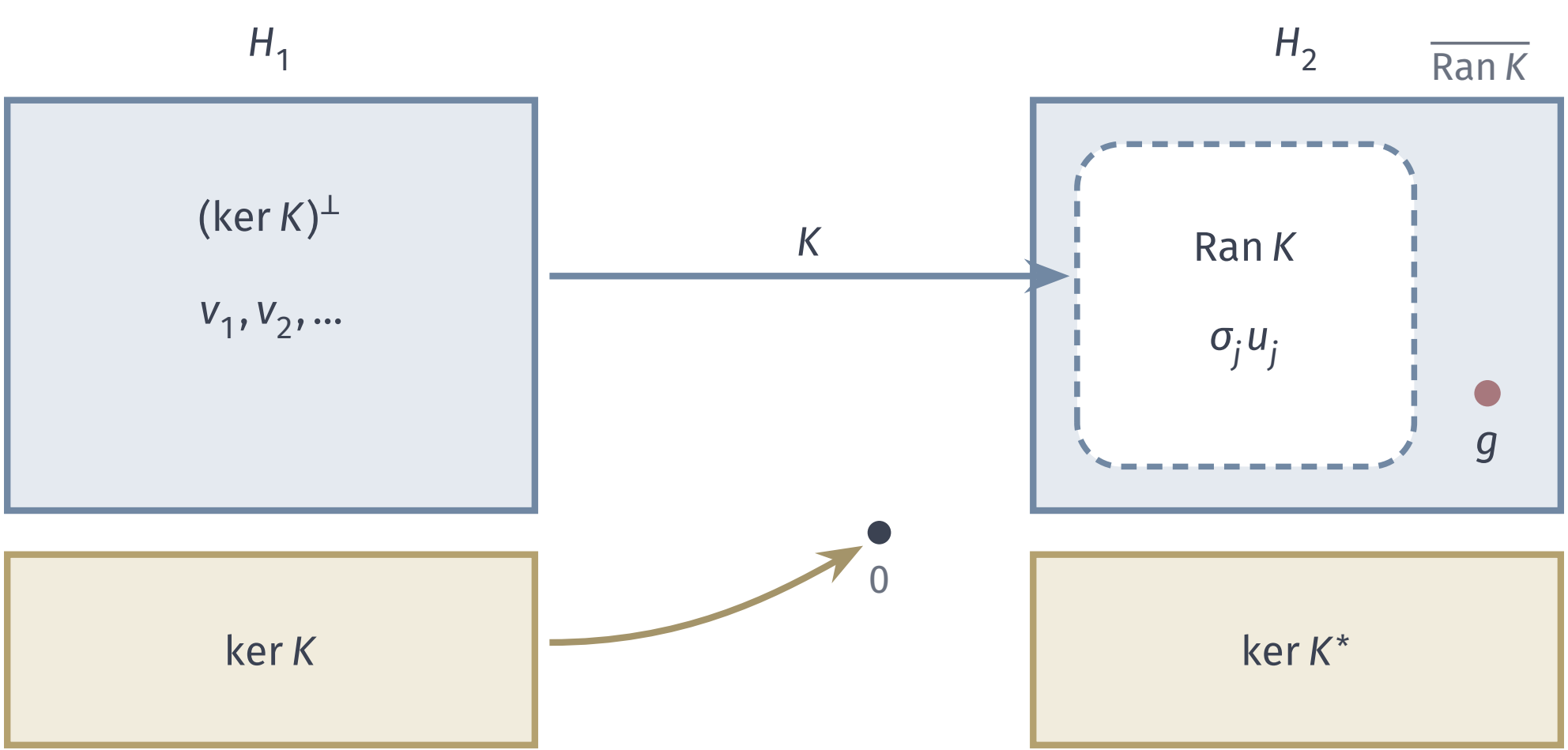
H_1
H_2
$\overline{\text{Ran } K}$
$(\ker K)^\perp$
$v_1, v_2, \ldots$
K
Ran K
$\sigma_j u_j$
g
0
$\ker K$
$\ker K^*$